

As for a vapour chamber, imagine the exact same thing as a heat pipe but the entire structure is a flat chamber. Vapour chambers are made of two plates of metal, in this case copper, which form a chamber with a spacer in it. The chamber has a material that acts as a wick which moves the liquid coolant that has condensed on the cold surface back towards the heat source. Vapour



Fins have a lot more going for them

chambers, are more difficult to manufacture but because they are flat, they ensure a very flush mating with the CPU. Also, the mounting pressure that can be applied to a vapour chamber is much greater than what can be applied to heat pipes. The reason we don't see vapour chambers in more CPU coolers is because they aren't cost effective. Heat pipes on the other hand are a lot cheaper to construct and incorporate. Also, a single vapour chamber, because of its large surface area has the same thermal conductivity as 2-4 heatpipes. This obviously depends on a lot of other factors as well, we're just providing a rough estimate.

Fins

Since the fins that are attached to the heat pipes are responsible for dissipating heat, there's a fair amount of R&D that goes into their design. In order to ensure that the air from the cooling fans passes over the fins, the ends of the fins are often bent to prevent air from leaking from the sides.



Channeling air for maximum cooling efficiency

The fin spacing is an important aspect as well. If the fins are too dense, then you need high static pressure fans which can apply more pressure and thus, force the air through the fins. Also, the stress on the fans increase which affects the longevity of the fans. If the static pressure is not enough then your cooling efficiency takes a hit. If the fins are spaced too far apart, then there wouldn't be enough surface area to dis-